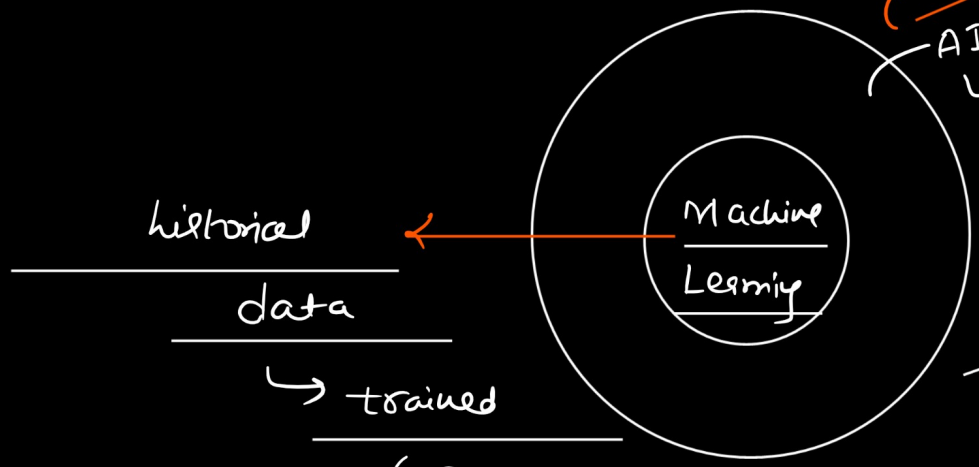
Machine Learning

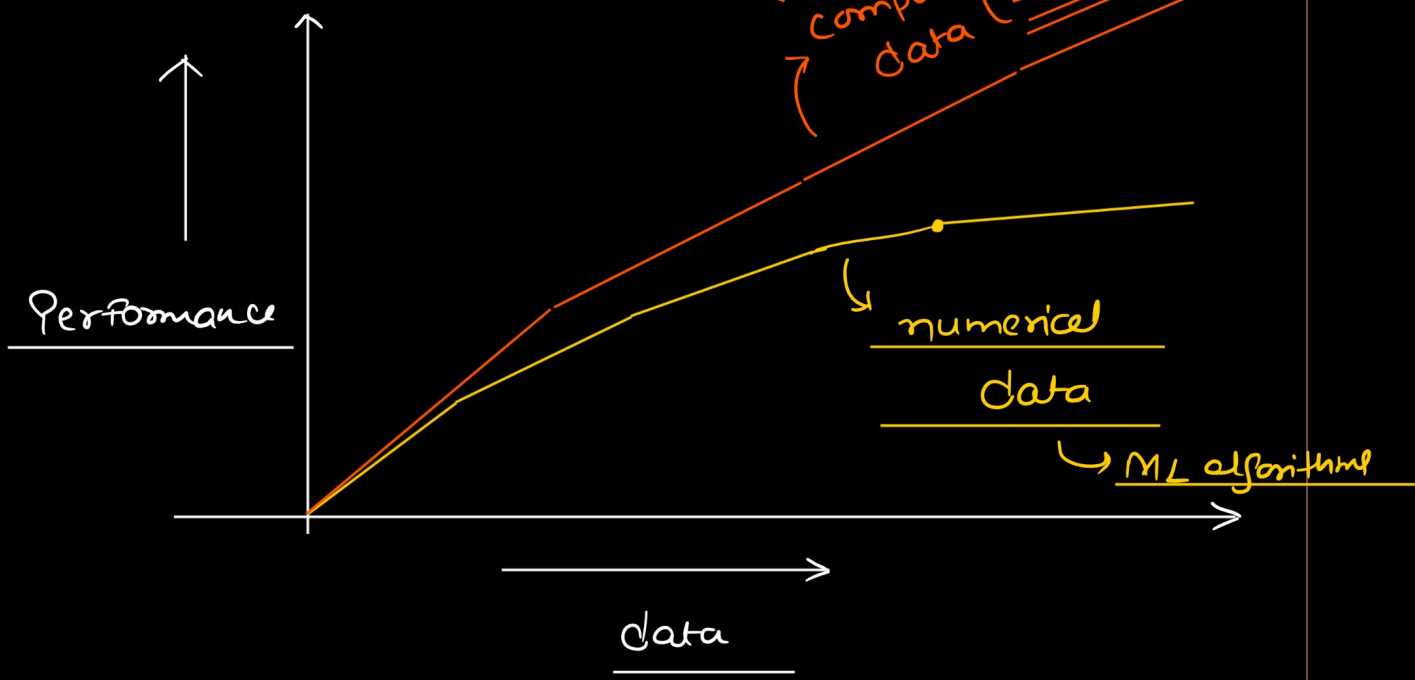


(self driving cars)

↳ End-End machine which has the ability to take decision as human beings

Deep learning

complex data (Image)



Why ML is still used in the industry??

- ↳ Data hungry
- ↳ Black box
- ↳ ML (easily interpretable)
- ↳ Resources ⟶ GPUs

Types of Machine Learning

Supervised Learning

learning from the labeled data

Outcome (included)

(Spam detection, diabetes data)

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0

Unsupervised Learning

find patterns in unlabelled data

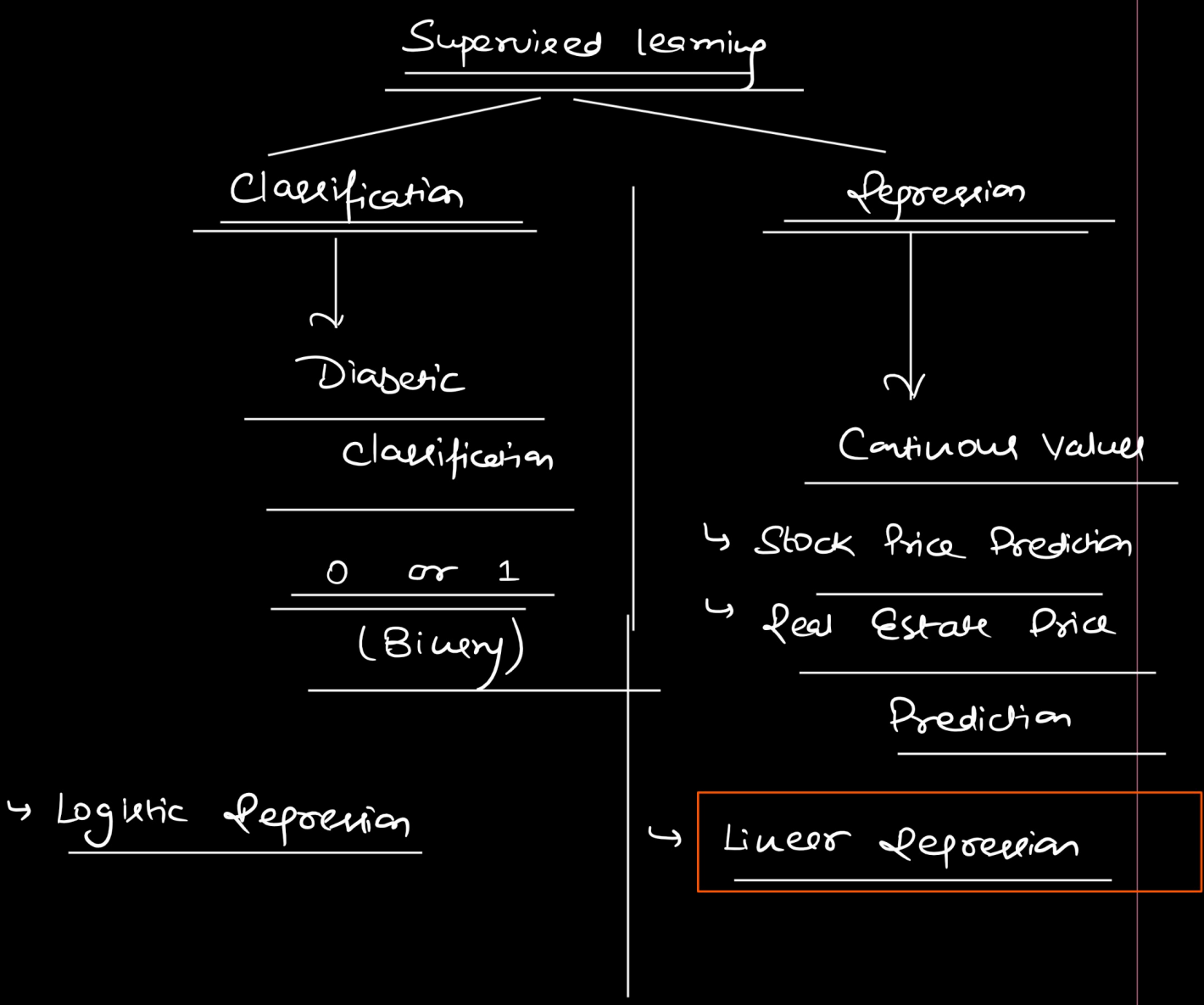
(clustering algorithm)

(Customer segmentation)

Reinforcement Learning

Learned through trial & error

Task → RLHF (Reinforcement Learning Human feedback)


Supervised learning
Classification
Regression
Diabetic
Classification
0 or 1
(Binary)
Continuous Values
Stock Price Prediction
Real Estate Price
Prediction
Logistic Regression
Linear Regression

Machine Learning Workflow

1. Business understanding → Experience / Domain knowledge
2. Data collection → DB (SQL)
3. Data Preparation → cleaning, feature (Exploratory Data Analysis) selection, outliers, missing values, Data Normalization
   (80%) → Quality Data
4. Choose a model → ML (numeric data)
   - Supervised
   - Unsupervised
5. Model training → Sklearn
6. Model Prediction
7. Evaluate the model
8. Deploy the model

Model

1. Maths behind the model
2. Implementation with public datasets

(UCI, Kaggle)